

A Top-Power-Form Obstruction for the Four-Vertex Path and Star Edge-Span Algebras

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Abstract

For the edge-span Chern algebras in ambient dimension four, we prove that the algebras of the four-vertex path and the four-vertex star are not isomorphic as unital graded $\mathbb{Q}$ -algebras. Their Hilbert series agree, so the obstruction uses multiplication. On the three-dimensional degree-one component, the ninth-power map is an intrinsic ternary nonic up to linear change of variables and scale. For the path it is a product of six linear forms; for the star it has three squared linear factors and an absolutely irreducible cubic factor. The calculation is exact over $\mathbb{Q}$ and is accompanied by two reproducible symbolic checks. The result is restricted to this pair and is presented as an alternative power-form certificate, not as a general reconstruction theorem.

1 Context and statement

Walters introduced edge-span Chern algebras and proved that they can be computed inside a finite polynomial quotient [Wal26b, Proposition 7.3 and equation (8)]. For trees, their Hilbert series depends only on the number of vertices and the ambient dimension, whereas in ambient dimension three the full algebra reconstructs the tree [Wal26b, Theorems 9.4 and 9.5]. The same paper explains that its reconstruction argument is specific to dimension three and asks about dimensions at least four [Wal26b, Remark 9.9].

A subsequent public working note in the author's repository describes a fifth-power-locus route intended to recover the line graph of every tree in ambient dimension four; the note expressly assigns the argument to future work and says that it still requires an independent audit [Wal26a]. Consequently, the pairwise conclusion below is not asserted as a priority claim. Its contribution is a different, pair-specific obstruction obtained from the factorization type of a single top power form. Exact multiplication-map certification also occurs in other graph-defined Artinian-algebra problems; for a nearby example concerning the weak Lefschetz property of whiskered-graph algebras, see [Zen26]. That work addresses a different algebra and does not support the theorem proved here.

Give every vertex variable algebraic degree one (half its cohomological degree). For a finite graph G , set

$$R_G^{(4)} = \frac{\mathbb{Q}[h_v : v \in V(G)]}{(h_v^4 : v \in V(G), \Delta_{uv} : uv \in E(G))}, \quad \Delta_{uv} = h_u^3 + h_u^2 h_v + h_u h_v^2 + h_v^3,$$

and let

$$A_G^{(4)} = \mathbb{Q}[h_u + h_v, h_u h_v : uv \in E(G)] \subseteq R_G^{(4)}.$$

The algebra is unital. Let $P = P_4$ have edges 12, 23, 34, and let $S = K_{1,3}$ have edges 12, 13, 14.

Theorem 1. *There is no unital degree-preserving $\mathbb{Q}$ -algebra isomorphism*

$$A_P^{(4)} \longrightarrow A_S^{(4)}.$$

This statement concerns abstract graded algebras: no preservation of the displayed edge generators or of vertex labels is assumed. It resolves only the specified four-vertex pair.

2 The intrinsic top power form

Lemma 2 (Top power form). *Let B be a finite-dimensional graded commutative $\mathbb{Q}$ -algebra with $\dim B_9 = 1$. Choose $0 \neq \omega \in B_9$ and define the homogeneous form $F_{B,\omega} \in \text{Sym}^9(B_1^\vee)$ by*

$$x^9 = F_{B,\omega}(x) \omega \quad (x \in B_1).$$

Under a graded algebra isomorphism, this form changes only by an invertible linear substitution and multiplication by a nonzero scalar. In particular, the degrees and multiplicities of its irreducible factors are invariants of the graded algebra.

Proof. Let $\phi : B \rightarrow C$ be a graded isomorphism, choose $0 \neq \eta \in C_9$, and write $\phi(\omega) = \lambda \eta$ with $\lambda \in \mathbb{Q}^\times$. Applying ϕ to the defining identity gives

$$F_{C,\eta}(\phi_1(x)) = \lambda F_{B,\omega}(x).$$

Here $\phi_1 \in \text{GL}(B_1, C_1)$. Its induced substitution is an isomorphism of polynomial unique-factorization domains, so it preserves irreducibility, factor degree, and multiplicity. $\square$

Specializing Walters's tree Hilbert-series formula to four vertices and ambient dimension four gives, for both P and S ,

$$(\dim A_d^{(4)})_{d=0}^9 = (1, 3, 9, 15, 21, 21, 17, 10, 4, 1). \quad (1)$$

Thus the three edge sums form a basis of degree one and the ninth graded piece is one-dimensional.

For the path, put

$$e_1 = h_1 + h_2, \quad e_2 = h_2 + h_3, \quad e_3 = h_3 + h_4, \quad \omega_P = h_1^2 h_2^2 h_3^2 h_4^3.$$

For the star, put

$$f_1 = h_1 + h_2, \quad f_2 = h_1 + h_3, \quad f_3 = h_1 + h_4, \quad \omega_S = h_2^3 h_3^3 h_4^3.$$

Proposition 3 (Exact top forms). *The two displayed top classes are nonzero. With respect to them, exact reduction in the defining quotients gives*

$$(c_1 e_1 + c_2 e_2 + c_3 e_3)^9 = 5040 c_1^2 c_2^2 c_3^2 (c_1 + c_2)(c_1 - c_3)(c_2 + c_3) \omega_P, \quad (2)$$

$$(c_1 f_1 + c_2 f_2 + c_3 f_3)^9 = -1680 c_1^2 c_2^2 c_3^2 Q(c_1, c_2, c_3) \omega_S, \quad (3)$$

where

$$\begin{aligned} Q(x, y, z) &= 3x^2 y + 3x^2 z + 3xy^2 + 8xyz + 3xz^2 + 3y^2 z + 3yz^2 \\ &= 3(x + y + z)(xy + xz + yz) - xyz. \end{aligned} \quad (4)$$

The exact certificate behind Proposition 3 is described in Section 4; no floating-point or finite-field inference is used.

3 The factorization obstruction

Lemma 4. *The cubic Q in (4) is absolutely irreducible in characteristic zero.*

Proof. The expression in (4) is symmetric. A reducible homogeneous cubic over an algebraic closure has a linear factor. After permuting the variables, the corresponding projective line may be written $z = ux + vy$. The coefficients of x^3, x^2y, xy^2, y^3 in $Q(x, y, ux + vy)$ are

$$3u(u+1), \quad 3u^2 + 6uv + 8u + 3v + 3, \quad 6uv + 3u + 3v^2 + 8v + 3, \quad 3v(v+1).$$

Vanishing of the first and last coefficients forces $u, v \in \{0, -1\}$. In the four cases

$$(u, v) = (0, 0), (0, -1), (-1, 0), (-1, -1),$$

the two middle coefficients are, respectively,

$$(3, 3), \quad (0, -2), \quad (-2, 0), \quad (1, 1).$$

None of these pairs vanishes. Hence Q has no linear factor over the algebraic closure and is absolutely irreducible. $\square$

Proof of Theorem 1. By (2), the path form has six distinct rational linear factors: c_1, c_2, c_3 have multiplicity two, and the other three have multiplicity one. By (3) and Lemma 4, the star form has three linear factors of multiplicity two and one irreducible cubic factor of multiplicity one. The factor-degree multisets are different. Lemma 2 therefore rules out an abstract graded $\mathbb{Q}$ -algebra isomorphism. $\square$

4 Exact certificate and reproducibility

The accompanying Python program `evidence/compute_exact.py` works over SymPy's exact rational domain. For each graph it performs the following finite calculation.

1. It computes a grevlex Gröbner basis of the defining ideal of $R_G^{(4)}$ and enumerates the standard monomials.
2. Degree by degree, it closes the unit under the three degree-one and three degree-two edge generators. Exact reduced row echelon form gives bases of the edge-span algebra and reproduces (1).
3. It exports every commutativity-reduced basis product of total degree at most nine (2,877 entries for each algebra). Iteration in this table computes the ninth powers.
4. Separately, it expands the ninth power multinomially in the ambient quotient and asserts equality with the multiplication-table result.

The output is `evidence/exact_algebra_certificate.json`. It records ambient Hilbert vector

$$(1, 4, 10, 17, 22, 22, 17, 10, 4, 1),$$

the nonzero degree-nine bases $\{\omega_P\}$ and $\{\omega_S\}$, the full multiplication tables, and the factorizations (2)–(3). The archived run used SymPy 1.14.0.

As a second implementation, `evidence/verify_top_forms.sing` works over the exact coefficient field $\mathbb{Q}(c_1, c_2, c_3)$. With Singular 4.4.1 it returns ambient quotient dimension 108 in both cases, nonzero normal forms for the chosen top monomials, and zero remainders for both asserted ninth-power identities. From the supplement root, the checks are invoked by

```
python3 evidence/compute_exact.py  
Singular evidence/verify_top_forms.sing
```

The Python command regenerates its JSON certificate. These exact computations establish Proposition 3; the intrinsic-invariance and irreducibility arguments are given independently in the text.

Scope and assistance disclosure. The theorem says nothing about other tree pairs or reconstruction in ambient dimension four. AI systems assisted with mathematical exploration, exact-code generation, verification, and editorial drafting. The displayed identities are exposed to reproducible rational-arithmetic checks, and the author takes responsibility for the statement, proof, citations, and scope.

References

- [Wal26a] Jackson Walters. Additional Experiments. Public working notes in the `graph-functor` repository, 2026. Accessed September 7, 2026.
- [Wal26b] Jackson Walters. Edge-Span Chern Algebras of Graphical Configuration Spaces. *arXiv preprint arXiv:2607.29483*, 2026.
- [Zen26] Zijian Zeng. The Whiskered-Graph Weak Lefschetz Conjecture for Graphs on Eight Vertices. SSRN working paper 7385138, 2026. Posted September 3, 2026.